where $r_p$ is the portfolio return. The unconstrained solution to both the CRRA utility and mean-variance utility problem with one risky asset and one risk-free asset paying $r_f$ is: [2]

$$x^* = \frac{1}{\gamma}\frac{\mu - r_f}{\sigma^2}, \tag{4.5}$$

where the asset has expected return $\mu$ and volatility $\sigma$. The investor holds $x^*$ in the risky asset and $(1 - x^*)$ in the risk-free asset. We developed this solution in detail in chapters 2 and 3. In this sense, CRRA and mean-variance utility are equivalent.

Since the optimal one-period weight is a mean-variance solution in equation (4.5), it nests as special cases many popular choices like equal weights, risk parity, market weights, and constant risk exposures considered in chapter 3. Thus, equation (4.5) does not necessarily mean a full-blown mean-variance solution—which we know does poorly. In the rest of the chapter, you can interpret the optimal one-period weight as any mean-variance optimal portfolio chosen by the investor in a single-period problem.

## 2.2. DYNAMIC PROGRAMMING

The dynamic portfolio choice problem is an *optimal control* problem. It is solved by *dynamic programming*—the same technique used to control nuclear power plants, send rockets to the moon, and value complicated derivative securities. (I admit, the last of these examples certainly feels much less impressive than the first two.) Portfolio choice turns out to be rocket science—literally. The reader not interested in the math behind representing long-run investing as a series of short-run investment problems can skip directly to section 2.3.

Long-horizon wealth is a product of one-period wealth:

$$W_{t+5} = W_t(1 + r_{p,t+1})(1 + r_{p,t+2}) \ldots (1 + r_{p,t+5}), \tag{4.6}$$

from equation (4.1), and we can apply CRRA utility (equation (4.3)) to each one-period wealth term. Apply CRRA expected utility to long-horizon wealth, we have a series of one-period CRRA utility problems:

$$E[U(W_{t+5})] = U(W_t)E\left[U(1 + r_{p,t+1})U(1 + r_{p,t+2}) \ldots U(1 + r_{p,t+5})\right]. \tag{4.7}$$

Since $U(W_t)$ appears outside the expectation in equation (4.7), it does not matter whether we start with \$1 or with \$1 million—the portfolio weights do not depend on the size of the initial wealth, which is the *wealth homogeneity* property

[2] Strictly speaking, the CRRA weight applies in continuous time or when the interval is very small. See Merton (1971).